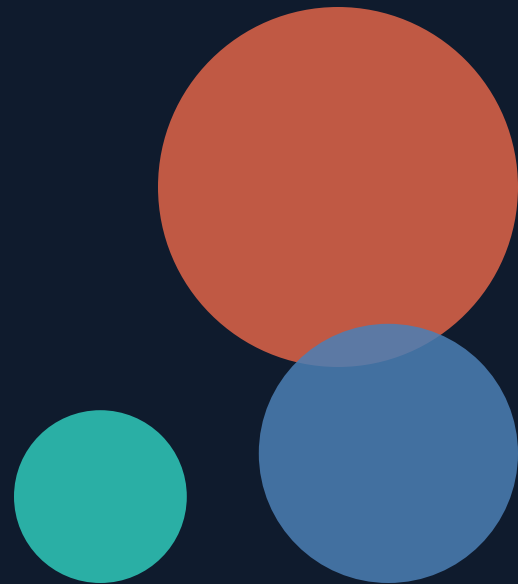


Ask or Act?

How AI Agent Autonomy Affects User Trust

Team A5 | BA830 Causal Inference | Spring 2026

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The Question

Should an AI assistant act first and tell you later,
or ask permission before every action?

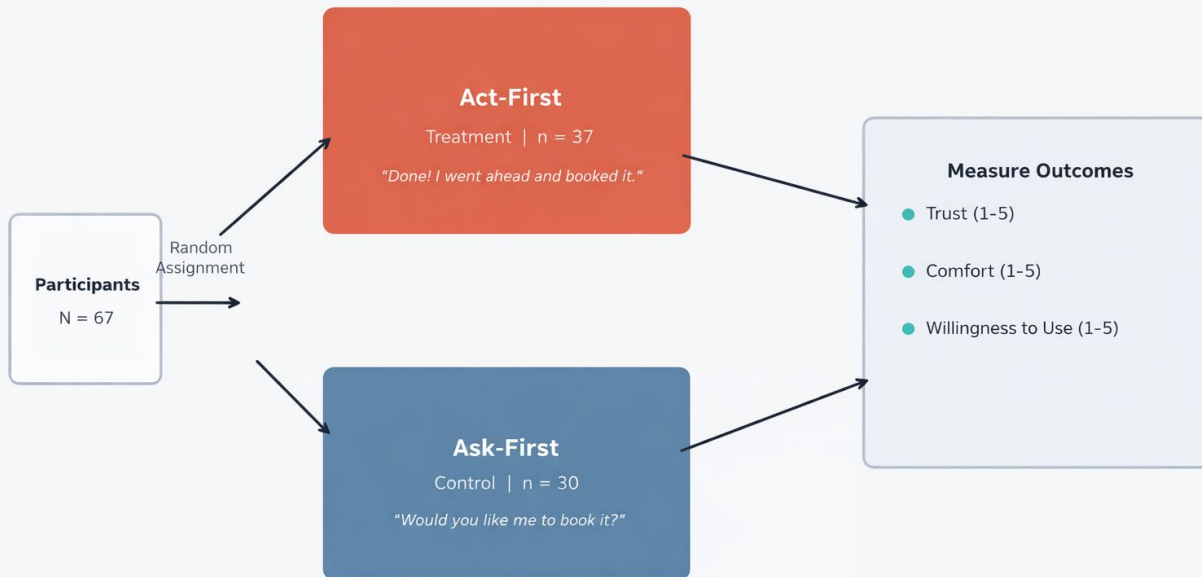
Act-First

*"Done! I went ahead and booked
Paseo for 7 PM Saturday."*

Ask-First

*"I found Paseo at 7 PM Saturday.
Would you like me to book it?"*

Experimental Design



What We Measured

Primary

Trust (1-5)
Comfort (1-5)

Secondary

Willingness to Use (1-5)

Covariates

Need for Control
Risk Aversion
Tech Comfort
Prior AI Experience

Scenario 1: Restaurant Booking

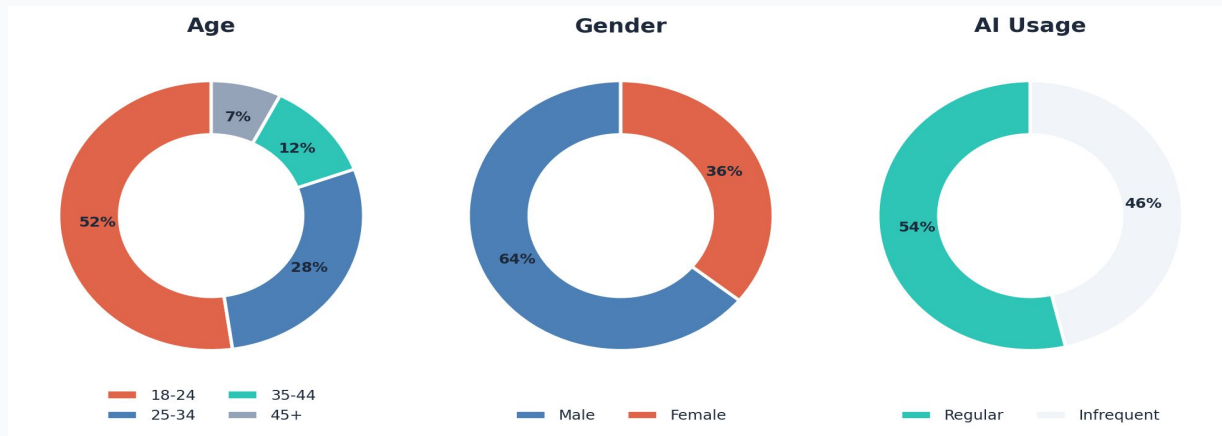
AI books a restaurant (or asks to book). Measures initial trust.

Who We Surveyed

67

participants

37 Act-First | 30 Ask-First

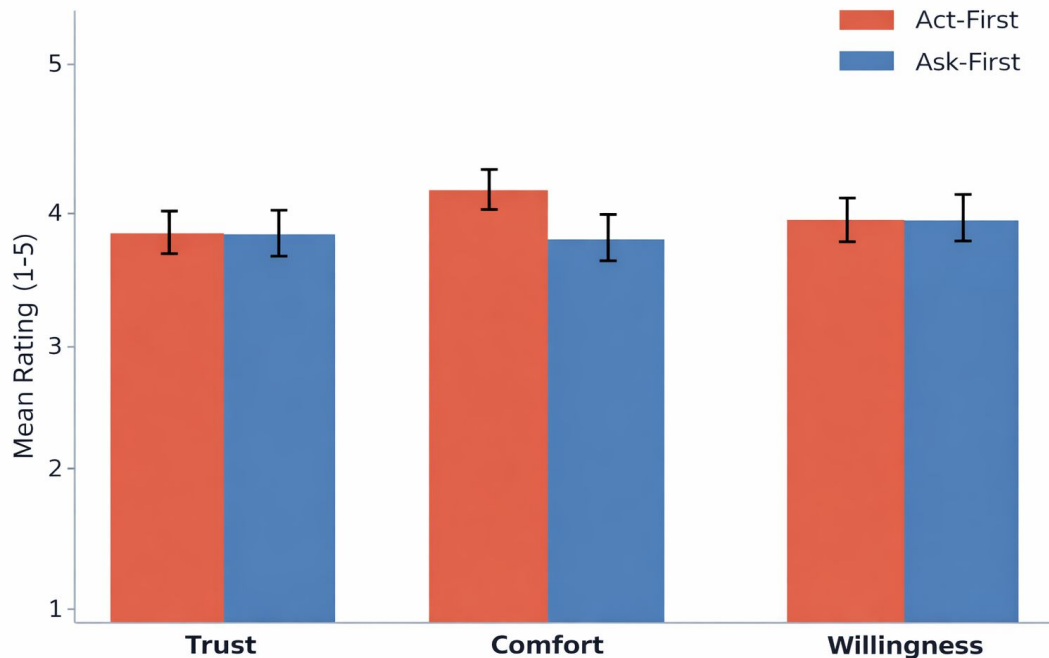


Balance Checks Passed

6 of 7 covariates balanced across conditions. Female showed marginal imbalance ($p = 0.090$) but does not predict outcomes, ruling out OVB.

Main Results

No significant difference in trust or willingness. Comfort shows a suggestive trend.



Trust

+0.007

$p = 0.978$

Comfort

+0.392

$p = 0.105$

Comfort (full model)

+0.412

$p = 0.089^*$

Willingness

-0.031

$p = 0.905$

The Comfort Signal

Comfort with Act-First

+0.41

points higher on 1-5 scale (full model)

p = 0.089

marginally significant at the 10% level

What This Means

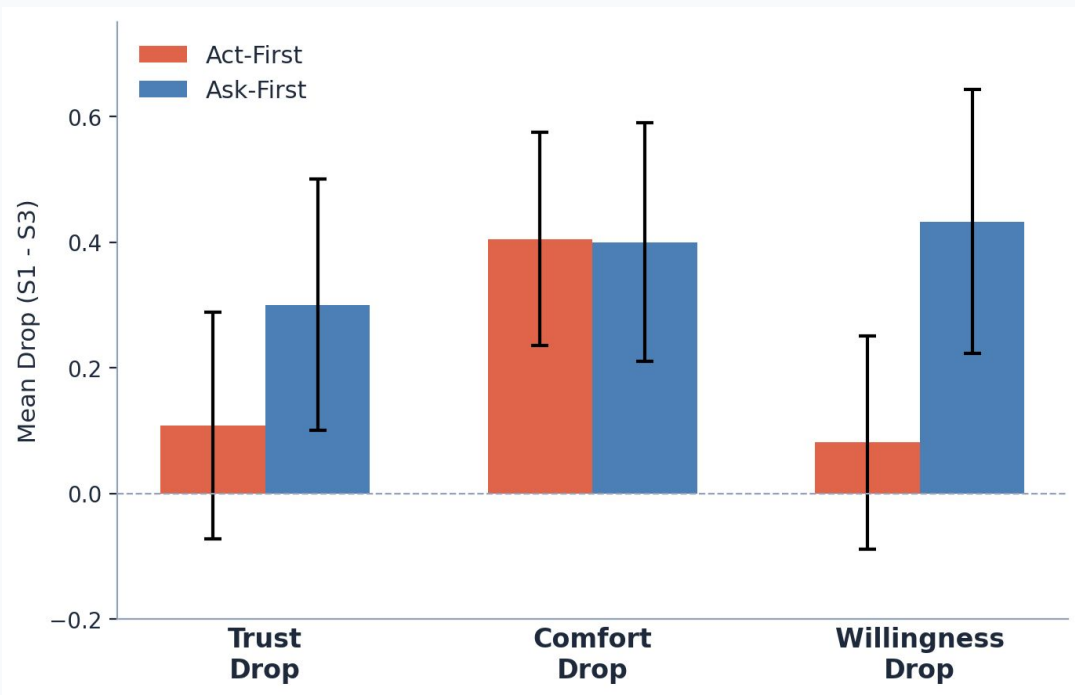
People who saw the AI act autonomously felt slightly **more comfortable** than those who were asked first.

This is the only outcome that approached statistical significance. The effect strengthened after controlling for covariates, suggesting potential confounding was masking a real (but small) treatment effect.

However, with $N = 67$, we are likely underpowered to detect effects this small with confidence.

When Things Go Wrong

Both conditions lose trust after an error. No difference in how much.



Trust Drop Difference

-0.19

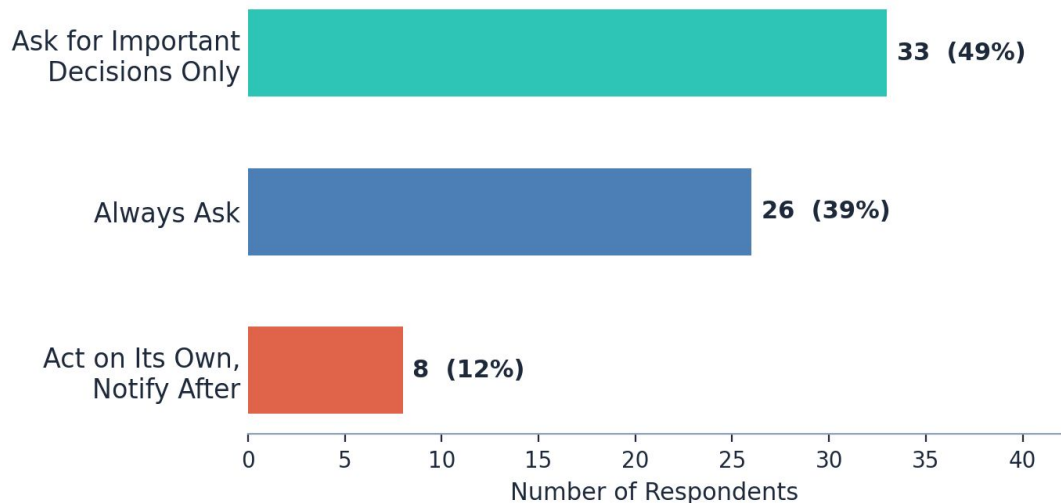
$p = 0.472$

Act-First users don't lose more trust after a mistake than Ask-First users.

Key Insight

Acting autonomously doesn't create a "betrayal effect." When AI makes a mistake, users react similarly regardless of whether it asked first.

What Users Actually Want



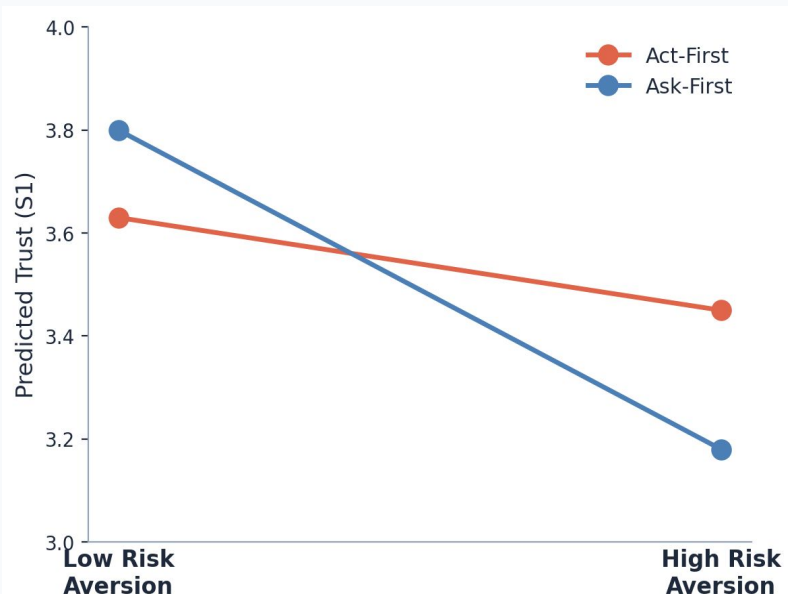
88%

want AI to ask before acting
(always or for important decisions)

Only 12% prefer full autonomy. Even when Act-First didn't hurt trust, users still want to be asked. The preference for control runs deeper than the outcome data suggests.

Who Responds Differently?

Heterogeneous treatment effects show suggestive patterns, though none reach significance.



Risk Aversion ($p = 0.305$)

Risk-averse users showed steeper trust declines under Ask-First, while Act-First users were more stable. Suggests high-risk-aversion users may paradoxically benefit from decisive AI.

Need for Control ($p = 0.909$) | Tech Comfort ($p = 0.490$)

Neither personality trait significantly moderated the treatment effect. Lines nearly parallel across both conditions.

Course Concepts

Concepts We Used

- **Randomization**

Qualtrics 50/50 random assignment ensures unbiased treatment allocation.

- **OLS with Covariates**

Controlling for NFC, RiskAversion, TechComfort, PriorAI to increase precision.

- **Balance Checks**

Verified covariate distributions are similar across groups.

- **OVb Analysis**

Checked if imbalanced covariates predict outcomes. They do not.

- **HTE**

Tested Condition x Covariate interactions to find differential effects.

Not Used (and Why)

- **Blocking**

Online survey with no natural blocking variable (e.g., lab section).

- **Clustering**

Individual-level randomization; no group-level assignment.

- **Failure to Treat**

Digital survey ensures 100% treatment delivery. No compliance issue.

Limitations

01 Manipulation Check

54% of participants failed the manipulation check, suggesting many didn't distinguish Act-First from Ask-First. Main analysis uses the full sample; robustness checks on the filtered subset ($N = 31$) show similar patterns.

02 Sample Size and Power

$N = 67$ is below our target of 80+. The comfort effect ($d = 0.40$) would need about 200 participants per group to detect at 80% power. Our study is underpowered for small effects.

03 External Validity

Convenience sample skews young, male, and tech-savvy. Hypothetical vignette is lower stakes than real AI interactions. Results may not generalize to older or less tech-comfortable populations.

Key Takeaway

**Acting first doesn't hurt.
But users still want to be asked.**

Trust and willingness are unaffected by autonomy level (all $p > 0.10$).

Comfort with Act-First trends higher (+0.41, $p = 0.089$) but needs more power.

88% of users prefer the AI to ask, regardless of outcome.

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Questions?